

Chapter 2

PHYSICAL LAYER AND MEDIA

Objectives

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- ▶ discusses the relationship between data, which are created by a device, and electromagnetic signals, which are transmitted over a medium.
- ▶ digital transmission. We discuss how we can convert digital or analog data to digital signals.
- ▶ analog transmission. We discuss how we can convert digital or analog data to analog signals.
- ▶ how we can use the available bandwidth efficiently. We discuss two separate, but related topics, multiplexing and spreading.
- ▶ characteristics of transmission media, both guided and unguided, in this chapter. Although transmission media operates under the physical layer, they are controlled by the physical layer.

Physical Layer

- ▶ interacts with the transmission media, the physical part of the network that connects network components together.
- ▶ One of the services provided by the physical layer is to create a signal that represents this stream of bits.
- ▶ The physical layer must also take care of the physical network, the **transmission medium**.
 - ▶ The transmission medium must be controlled by the physical layer. The physical layer decides on the directions of data flow. The physical layer decides on the number of logical channels for transporting data coming from different sources.

Note

To be transmitted, data must be transformed to electromagnetic signals.

3-1 ANALOG AND DIGITAL

Data can be analog or digital. The term **analog data** refers to information that is continuous; **digital data** refers to information that has discrete states. Analog data take on continuous values. Digital data take on discrete values.

Topics discussed in this section:

- Analog and Digital Data
- Analog and Digital Signals
- Periodic and Nonperiodic Signals



Analog and Digital Data

- Data can be analog or digital.
- Analog data are continuous and take continuous values.
- Digital data have discrete states and take discrete values.

Analog and Digital Signals

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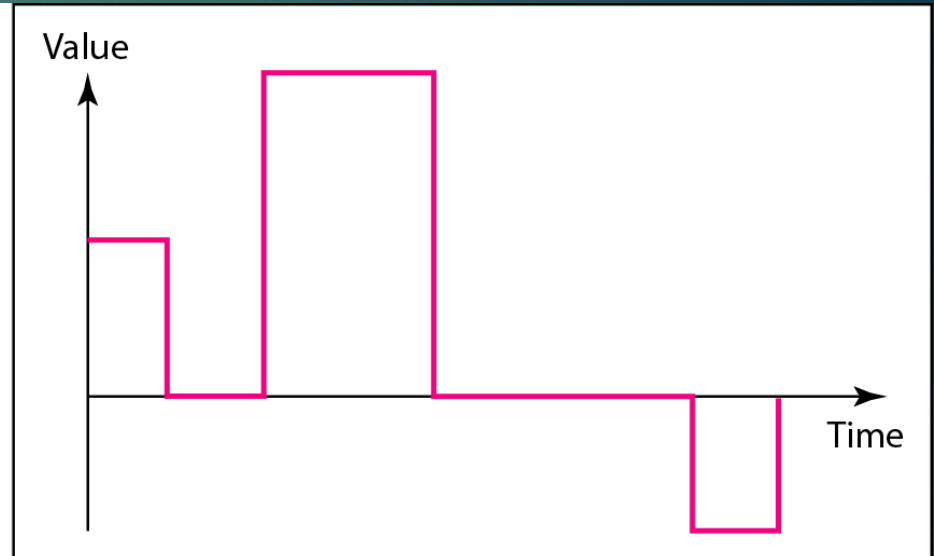
- Signals can be analog or digital.
- Analog signals can have an infinite number of values in a range.
- Digital signals can have only a limited number of values.

Figure 3.1 Comparison of analog and digital signals

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a. Analog signal



b. Digital signal

Periodic and Nonperiodic Signals

- Both analog and digital signals can take one of two forms: *periodic* or *nonperiodic*
- A **periodic** signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a cycle.
- A **nonperiodic** signal changes without exhibiting a pattern or cycle that repeats over time.

3-2 PERIODIC ANALOG SIGNALS

In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

Periodic analog signals can be classified as **simple** or **composite**. A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

Topics discussed in this section:

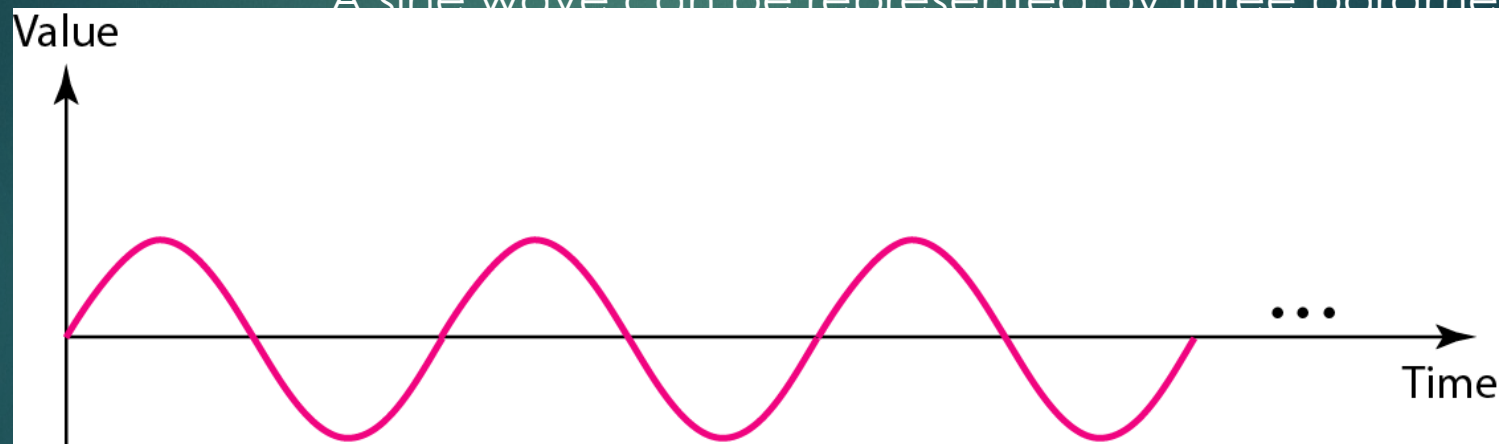
- Sine Wave
- Wavelength
- Time and Frequency Domain
- Composite Signals
- Bandwidth

Figure 3.2 A sine wave

simple oscillating curve, its change over the course of a cycle is smooth and consistent, a continuous, rolling flow.

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A sine wave can be represented by three parameters: amplitude, frequency, and phase.



periodic analog signal.

Signal characteristics

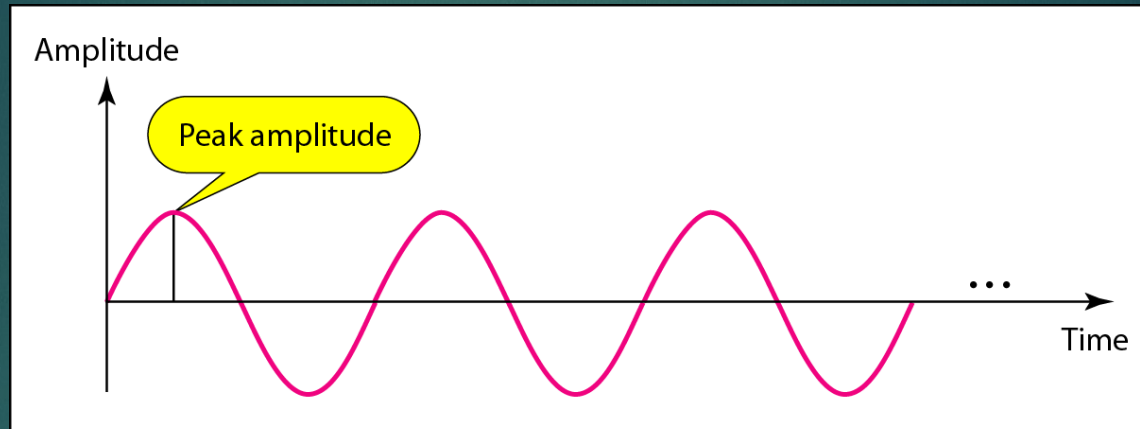
The peak amplitude

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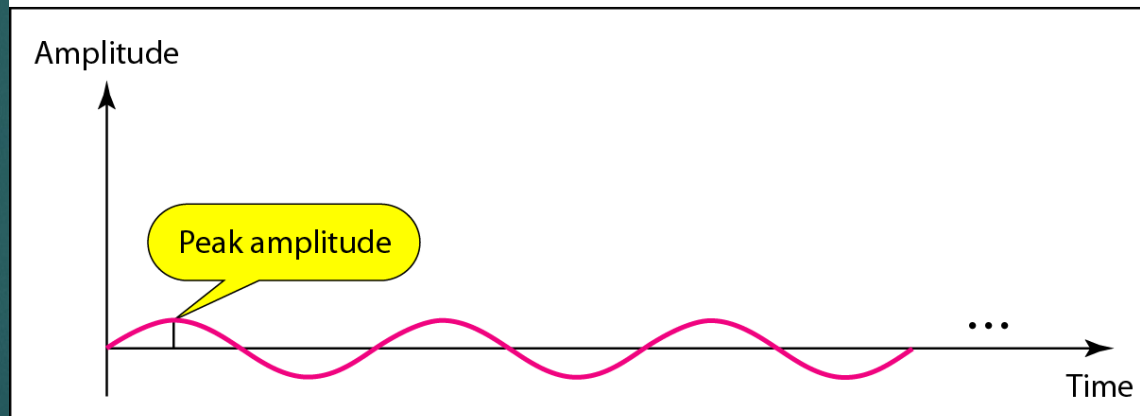
- The peak amplitude of a signal is the absolute value of its highest intensity, proportional to the energy it carries. For electric signals, peak amplitude is normally measured in ***volts***.

Figure 3.3 Two signals with the same phase and frequency,
but different amplitudes

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a. A signal with high peak amplitude



b. A signal with low peak amplitude

- ***Period and Frequency***

Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

Frequency refers to the number of periods in 1 s.

Note

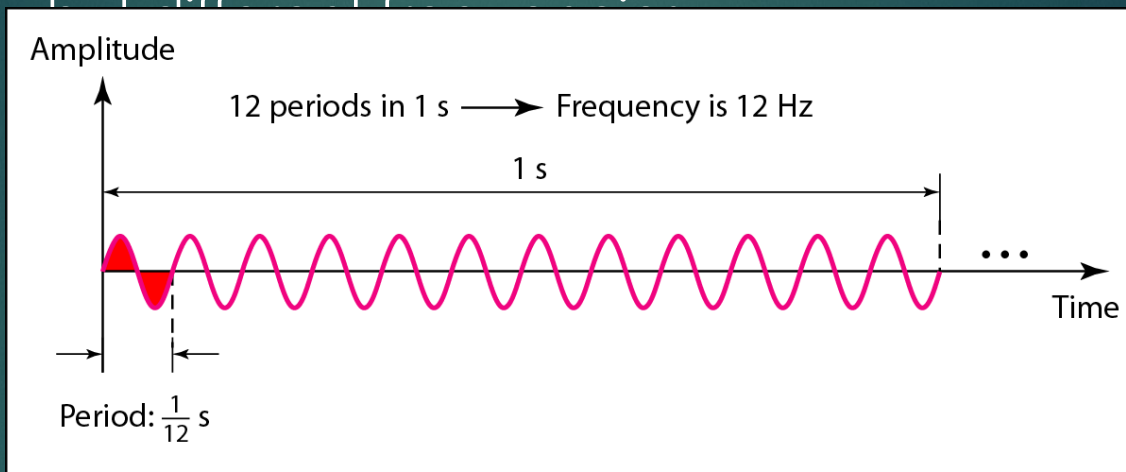
Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

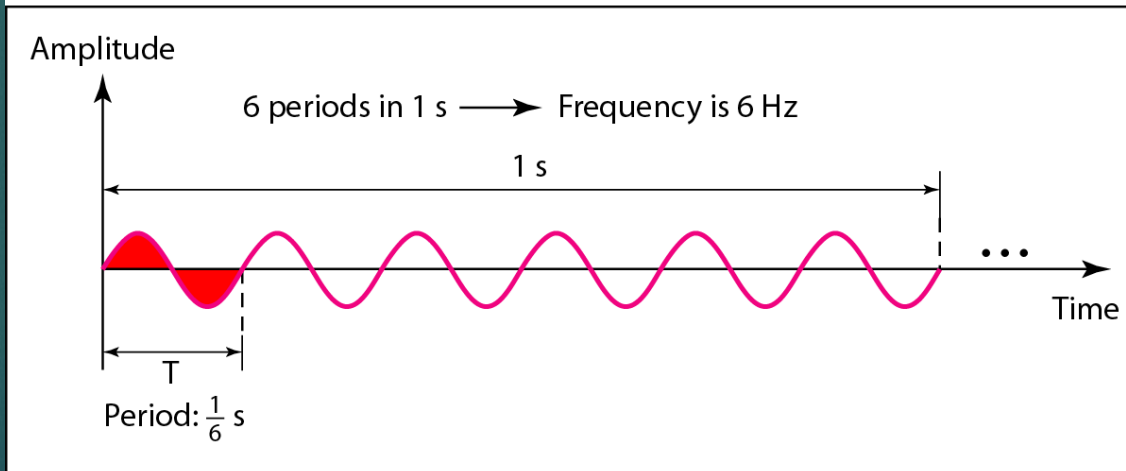
Figure 3.4 Two signals with the same amplitude and phase,

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a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Period is formally expressed in seconds. Frequency is formally expressed in Hertz (Hz), which is cycle per second.

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Table 3.1 Units of period and

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Frequency

- Frequency is the rate of change with respect to time.
- Change in a short span of time means high frequency.
- Change over a long span of time means low frequency.



Note

If a signal does not change at all, its frequency is zero.
If a signal changes instantaneously, its frequency is infinite.



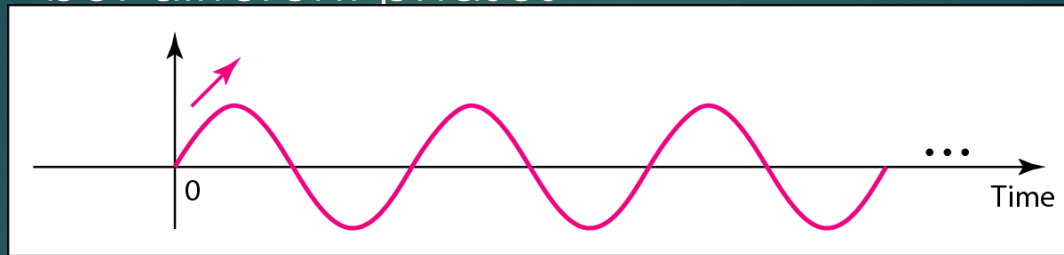
Note

Phase describes the position of the waveform relative to time 0.

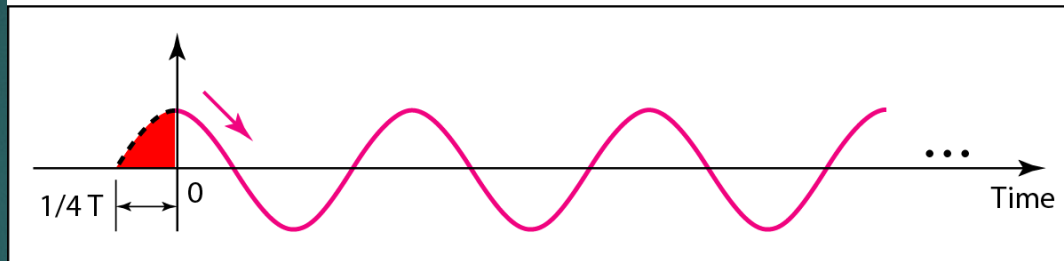
Figure 3.5 Three sine waves with the same amplitude and frequency, but different phases

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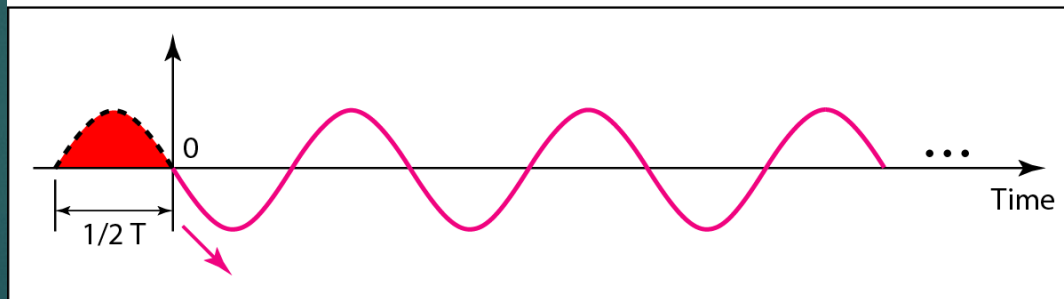
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a. 0 degrees



b. 90 degrees



c. 180 degrees



Example

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A sine wave is offset 1/6 cycle with respect to time 0. What is its phase in degrees and radians?

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Solution

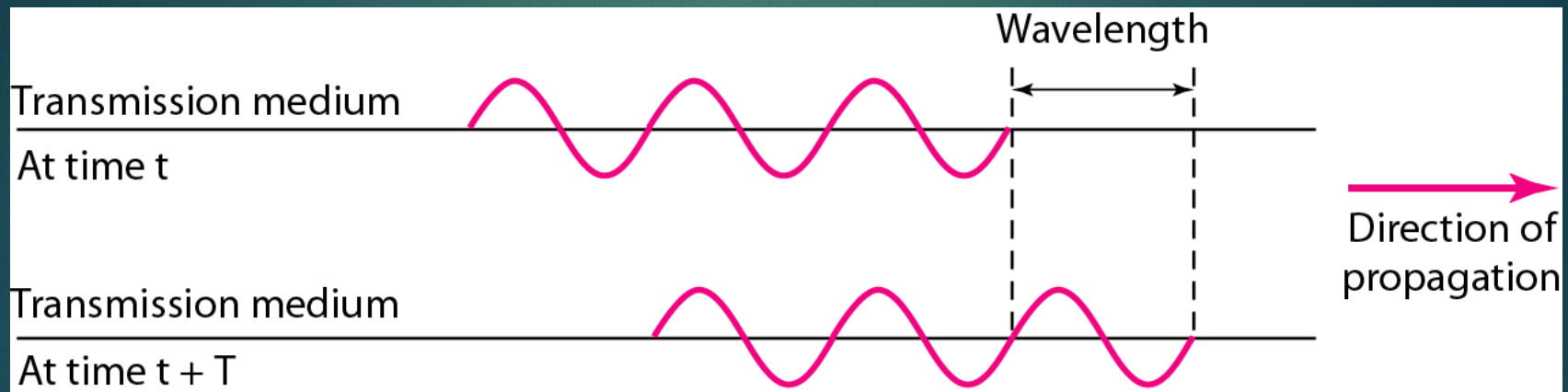
We know that 1 complete cycle is 360° . Therefore, 1/6 cycle is

$$\frac{1}{6} \times 360 = 60^\circ = 60 \times \frac{2\pi}{360} \text{ rad} = \frac{\pi}{3} \text{ rad} = 1.046 \text{ rad}$$

- Wavelength is a property of any type of signal.
- The wavelength is the distance a simple signal can travel in one period.
- Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium.
- frequency of a signal is independent of the medium, the wavelength depends on both the frequency and the medium.
- In data communications, we often use wavelength to describe the transmission of light in an optical fiber.

Figure 3.6 Wavelength and period

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Wavelength can be calculated if one is given the propagation speed (the speed of light) and the period of the signal. However, since period and frequency are related to

each other, if we represent wavelength by λ , propagation speed by c (speed of light), and frequency by f , we get .

Wavelength = propagation speed x period of a signal =
propagation speed / frequency

The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal.

For example, in a vacuum, light is propagated with a speed of 3×10^8 m/s. That speed is lower in air and even lower in cable.

The wavelength is normally measured in micrometers (microns) instead of meters.

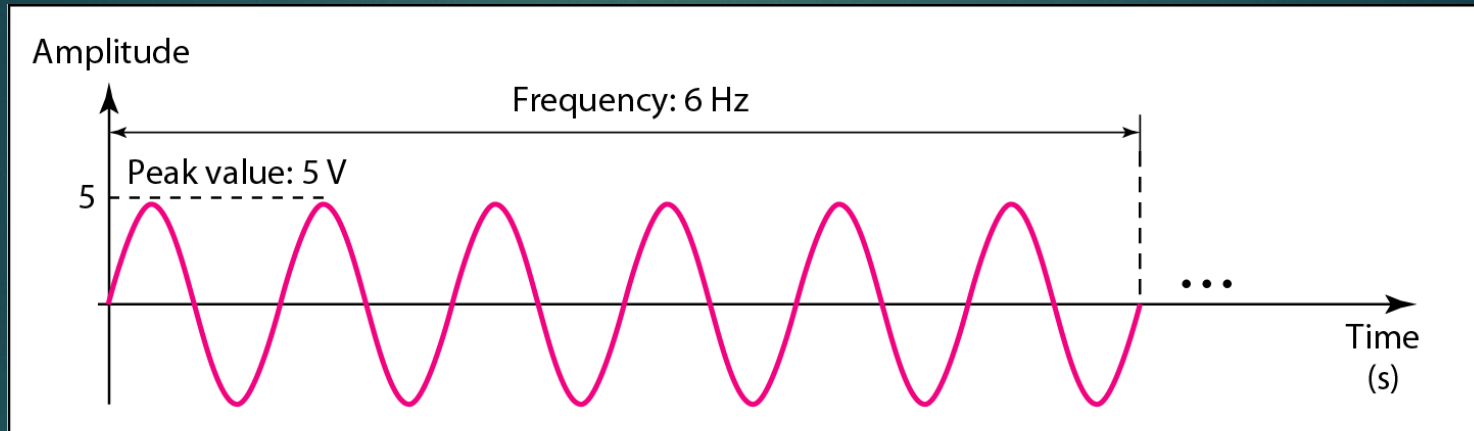
For example, the wavelength of a signal with a frequency of 4×10^{14} Hz in air is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{4 \times 10^{14}} = 0.75 \times 10^{-6} \text{ m} = 0.75 \mu\text{m}$$

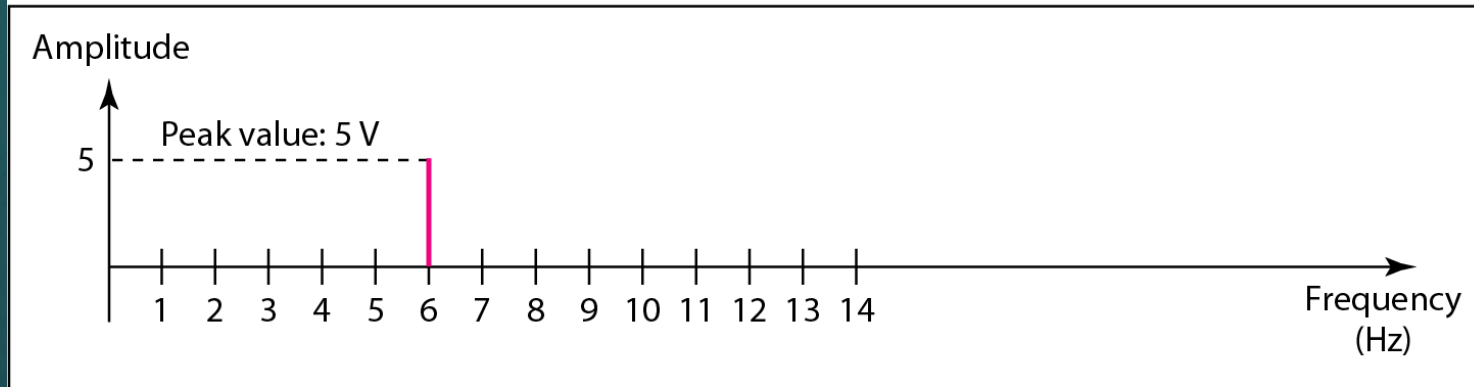
(0.5 μm)

In a coaxial or fiber-optic cable, however, the wavelength is shorter because the

Figure 3.7 The time-domain and frequency-domain plots of a sine wave



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



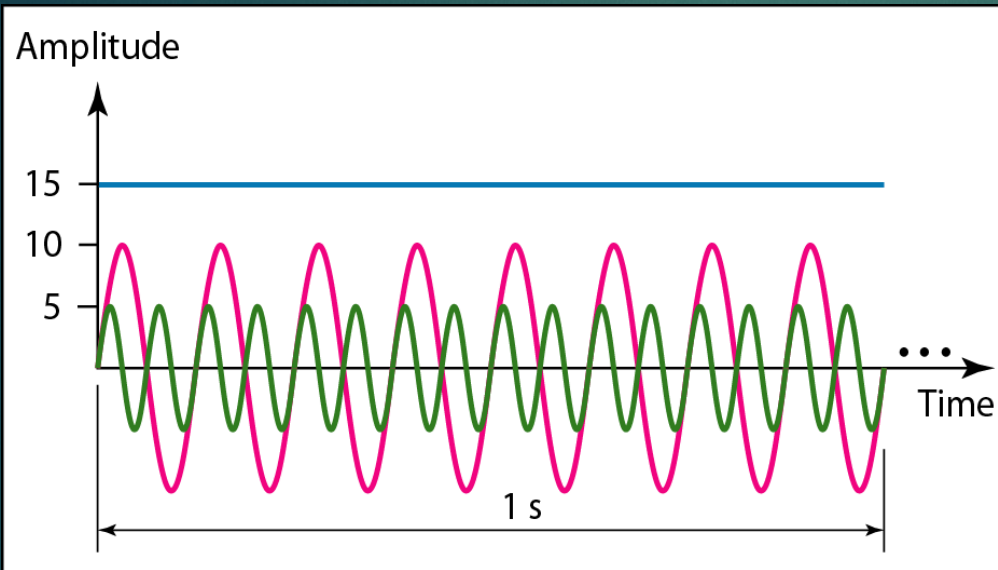
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Note

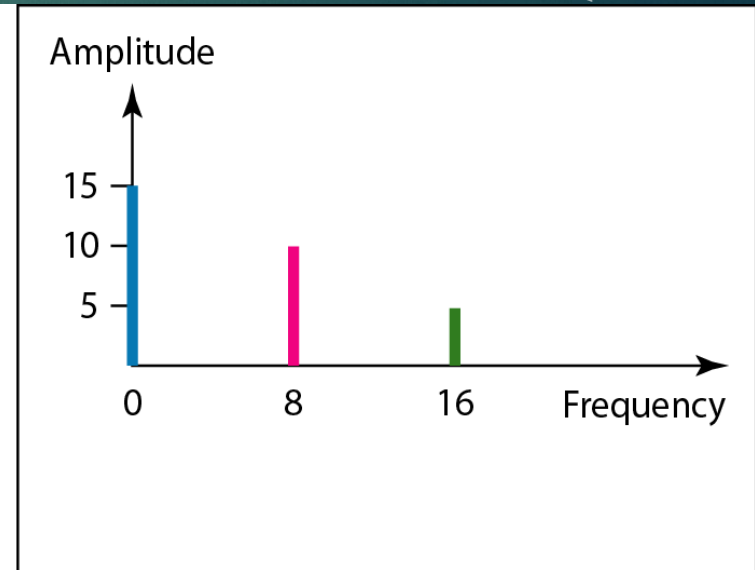
A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

Figure 3.8 The time domain and frequency domain of three sine waves

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a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

Signals and Communication

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- ▶ A single-frequency sine wave is not useful in data communications
- ▶ We need to send a composite signal, a signal made of many simple sine waves.
- ▶ According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.

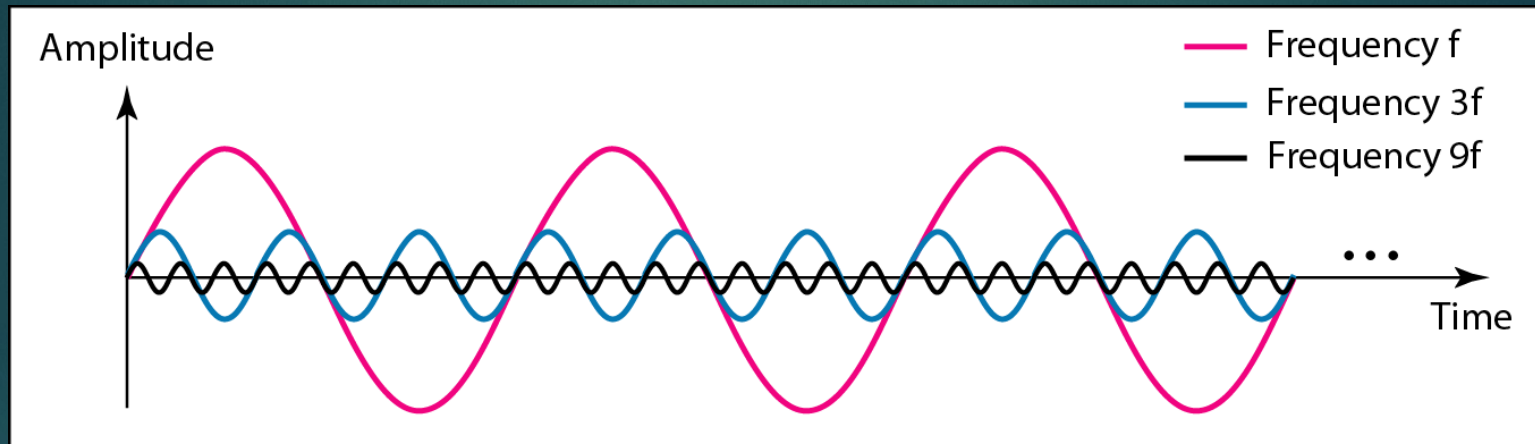
Composite Signals and Periodicity

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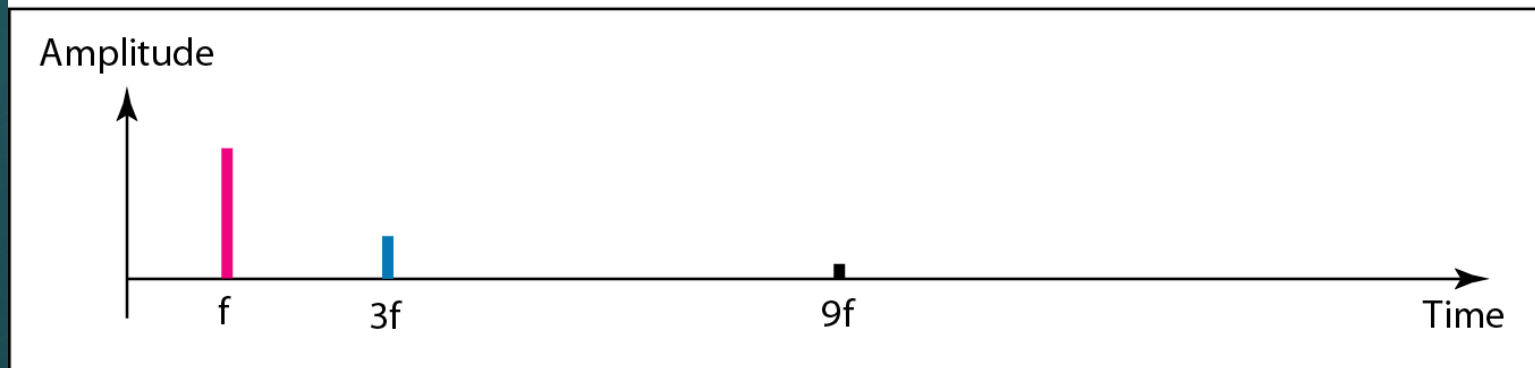
- ▶ If the composite signal is **periodic**, the decomposition gives a series of signals with **discrete** frequencies.
- ▶ If the composite signal is **nonperiodic**, the decomposition gives a combination of sine waves with **continuous** frequencies.

Figure 3.10 Decomposition of a composite periodic signal in the time and frequency domains

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a. Time-domain decomposition of a composite signal



b. Frequency-domain decomposition of the composite signal

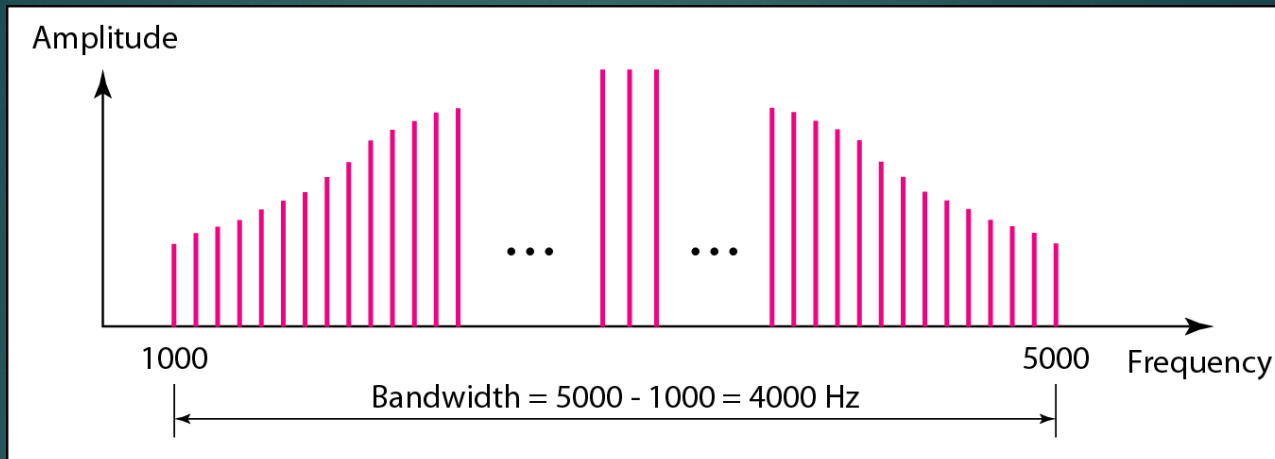
Bandwidth and Signal Frequency

- ▶ The bandwidth of a composite signal is the **difference** between the highest and the lowest frequencies contained in that signal.

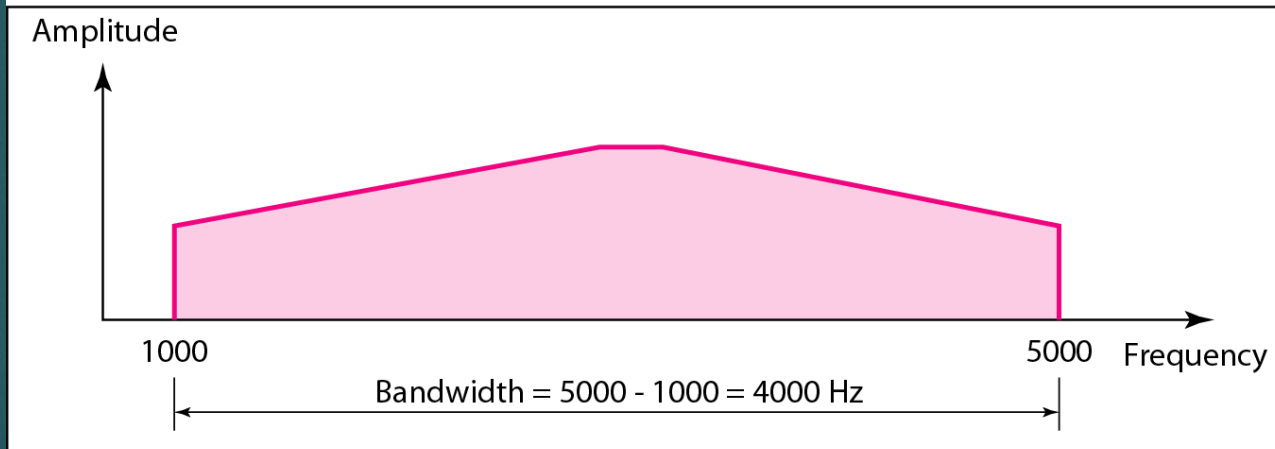
Figure 3.12 The bandwidth of periodic and nonperiodic composite signals

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a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

3-3 DIGITAL SIGNALS

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In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

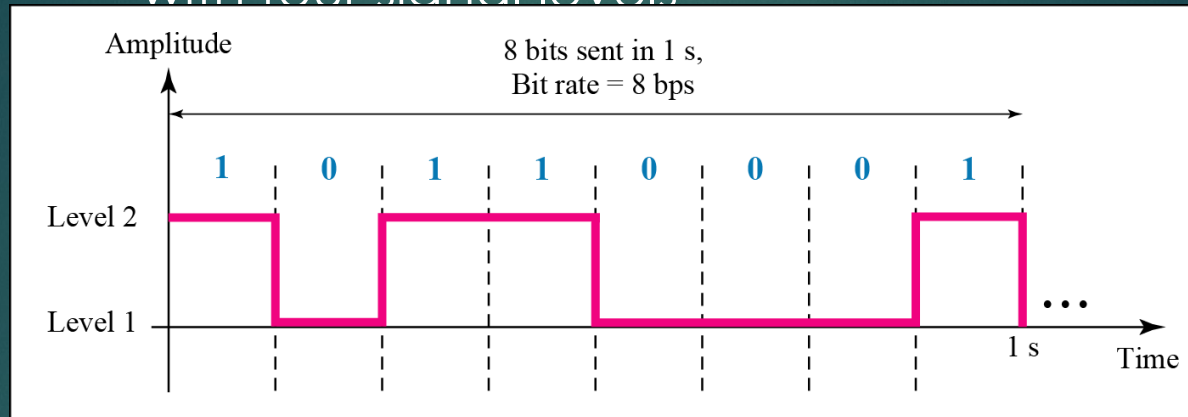
Topics discussed in this section:

- Bit Rate
- Bit Length
- Digital Signal as a Composite Analog Signal
- Application Layer

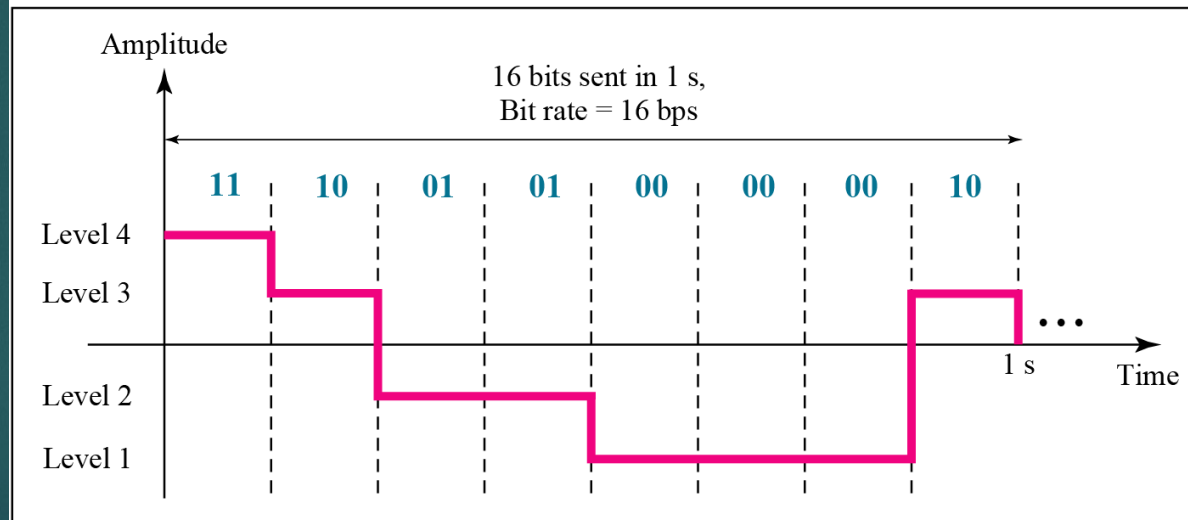
Figure 3.16 Two digital signals: one with two signal levels and the other

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with four signal levels



a. A digital signal with two levels



b. A digital signal with four levels

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Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics. Another *term-bit rate* (instead of *frequency*)-is used to describe digital signals.

The **bit rate** is the number of bits sent in 1s, expressed in bits per second (bps). Figure 3.16 shows the bit rate for two signals.

We discussed the concept of the wavelength for an analog signal: the distance one cycle occupies on the transmission medium. We can define something similar for a digital signal: the bit length. The bit length is the distance one bit occupies on the transmission medium.

Bit length = propagation speed x bit duration



Bit Length

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Data Rate Limits: one of the most important consideration in data communications is how fast we can send data, in bits per second over a channel. Data rate depends on three factors:

- a. The bandwidth available
- b. The level of the signals we use
- c. The quality of the channel (the level of noise)

3-4 TRANSMISSION IMPAIRMENT

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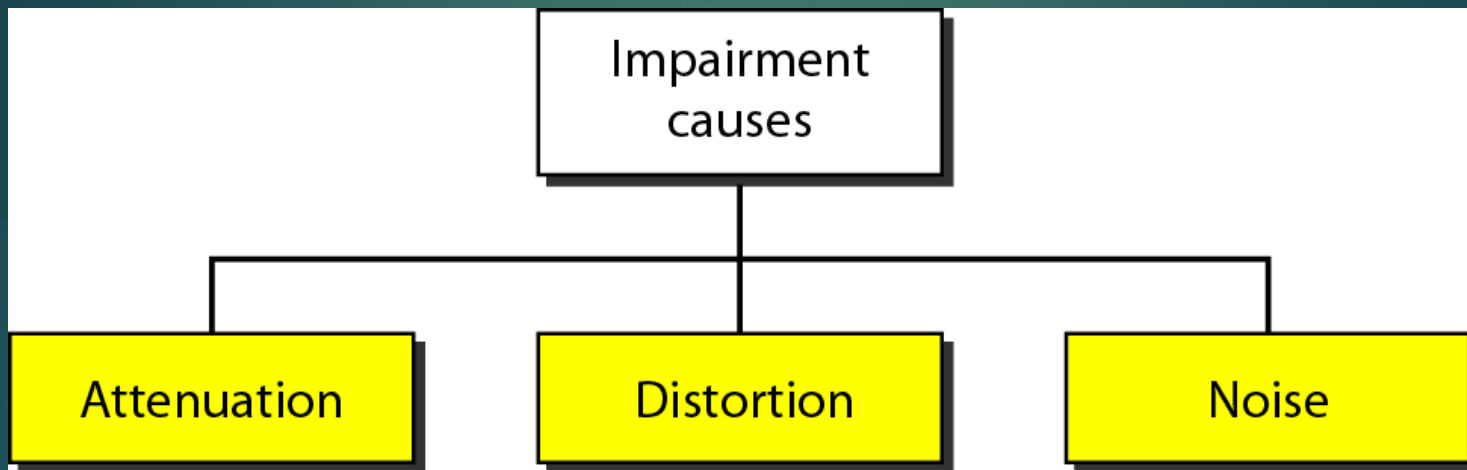
Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are **attenuation**, **distortion**, and **noise**.

Topics discussed in this section:

- Attenuation
- Distortion
- Noise

Figure 3.25 Causes of impairment

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Attenuation

- ▶ where signal strength falls off with distance
- ▶ depends on medium
- ▶ received signal strength must be:
 - ▶ strong enough to be detected
 - ▶ sufficiently higher than noise to receive without error
- ▶ so increase strength using amplifiers/repeaters
- ▶ is also an increasing function of frequency
- ▶ so equalize attenuation across band of frequencies used
 - ▶ eg. using loading coils or amplifiers

Measurement of Attenuation

- ▶ Attenuation is measured in terms of Decibels.
- ▶ The decibel (dB) measures the relative strengths of two signals or one signal at two different points. Note that the decibel is negative if a signal is attenuated and positive if a signal is amplified.

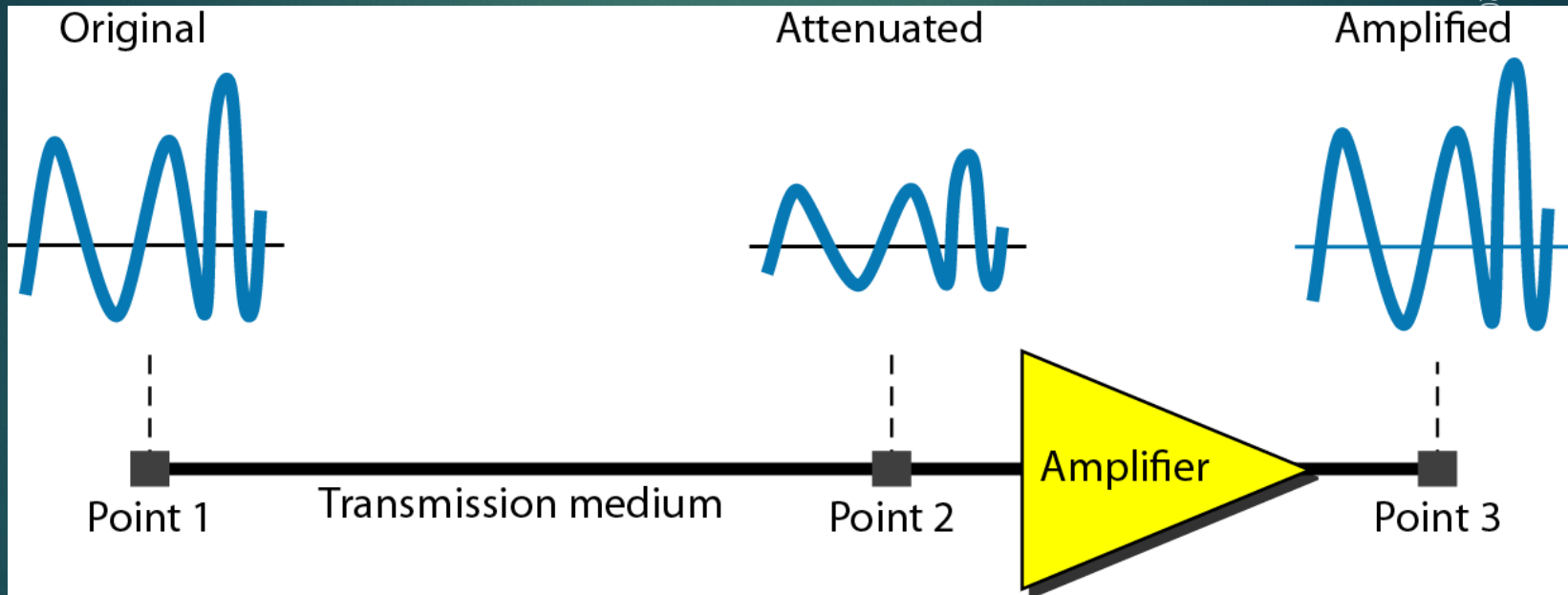
$$\text{dB} = 10\log_{10} P_2/P_1$$

P_1 - input signal

P_2 - output signal

Figure 3.26
Attenuation

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Distortion

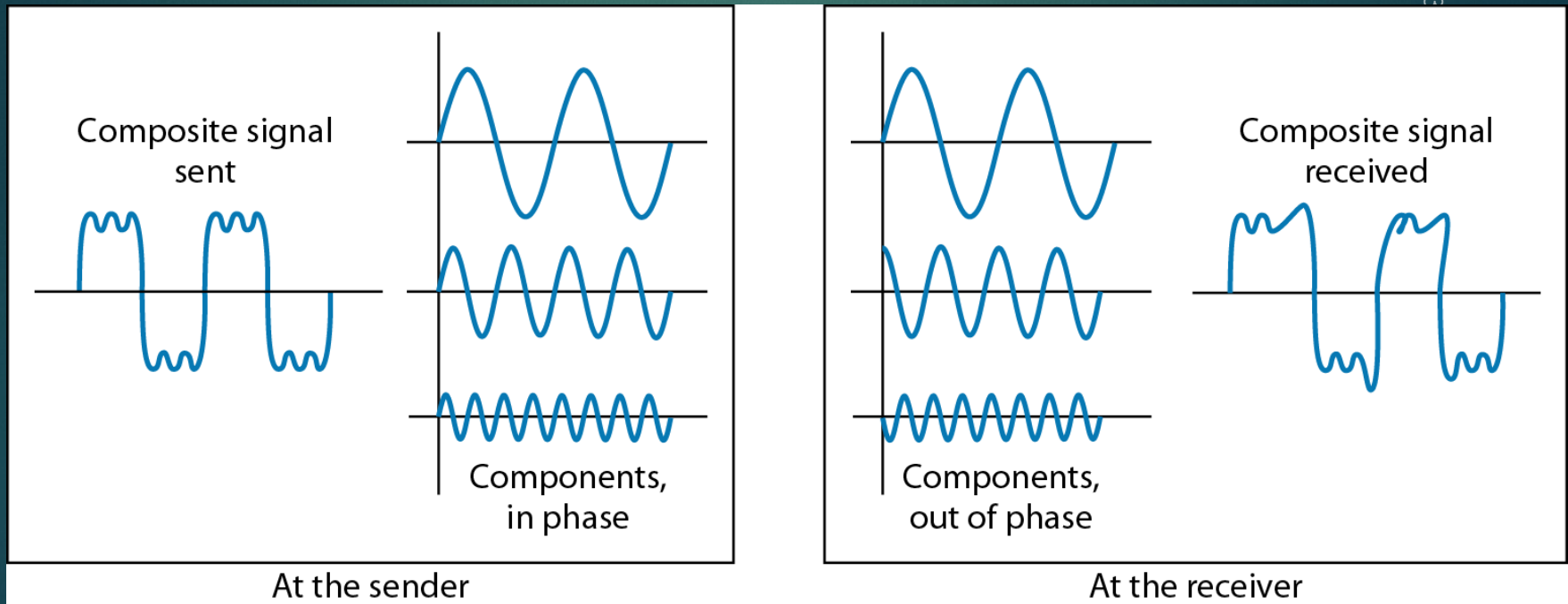
- ▶ Means that the signal changes its form or shape
- ▶ Distortion can occur in a composite signal made of different frequencies.
- ▶ Each frequency component has its own **propagation speed** traveling through a medium.
- ▶ The different components therefore arrive with **different delays** at the receiver.
- ▶ That means that the signals have **different phases** at the receiver than they did at the source.

Figure 3.28

Distortion

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Noise

- ▶ Noise is any unwanted signal that is mixed or combined with the original signal during transmission.
- ▶ Due to noise the original signal is altered and signal received is not same as the one sent.

Noise

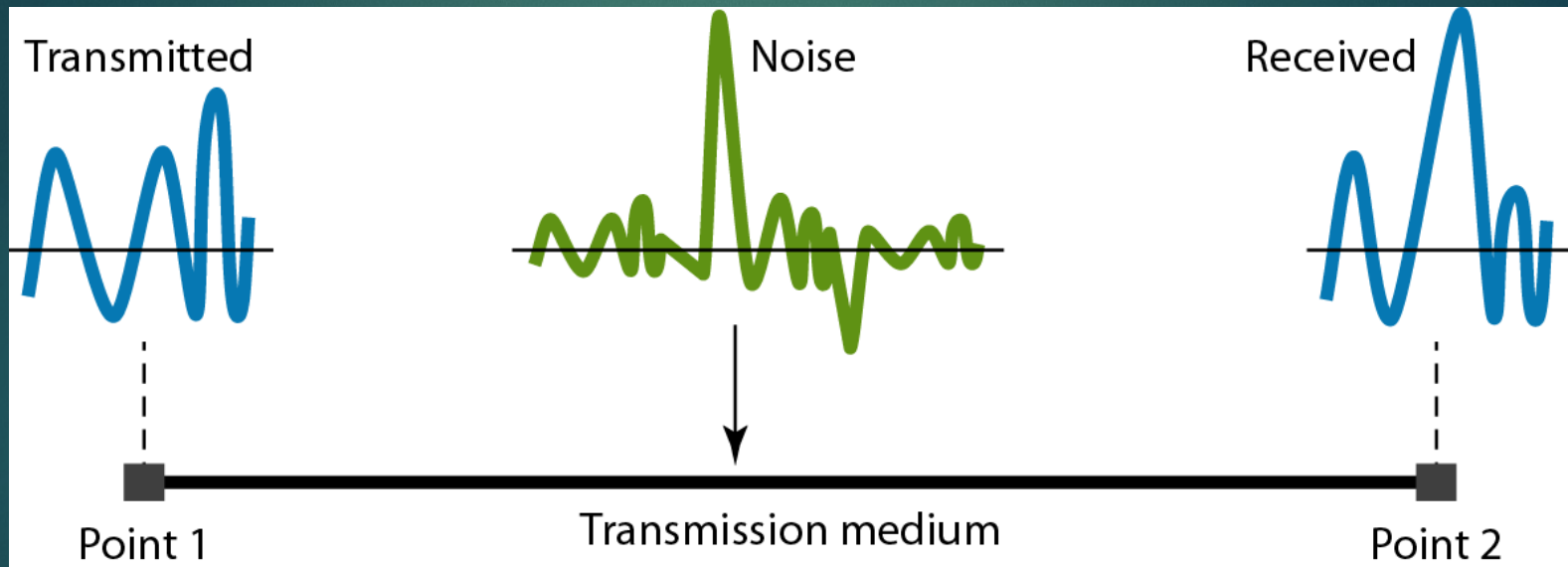
- ▶ There are different types of noise
 - ▶ **Thermal** - is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
 - ▶ **Induced** – Induced noise comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna. .
 - ▶ **Crosstalk** - Crosstalk is the effect of on the other. One wire acts as a sending antenna and the other as the receiving antenna.
 - ▶ **Impulse** - Impulse noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

Figure 3.29

Noise

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Signal to Noise Ratio (SNR)

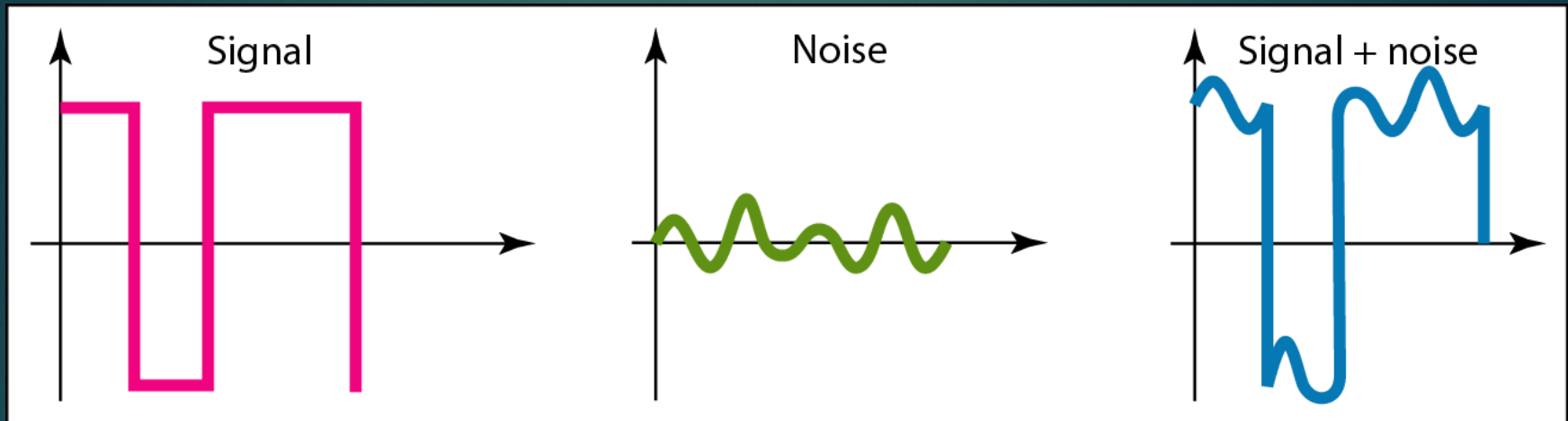
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- ▶ To measure the quality of a system the SNR is often used. It indicates the strength of the signal wrt the noise power in the system.

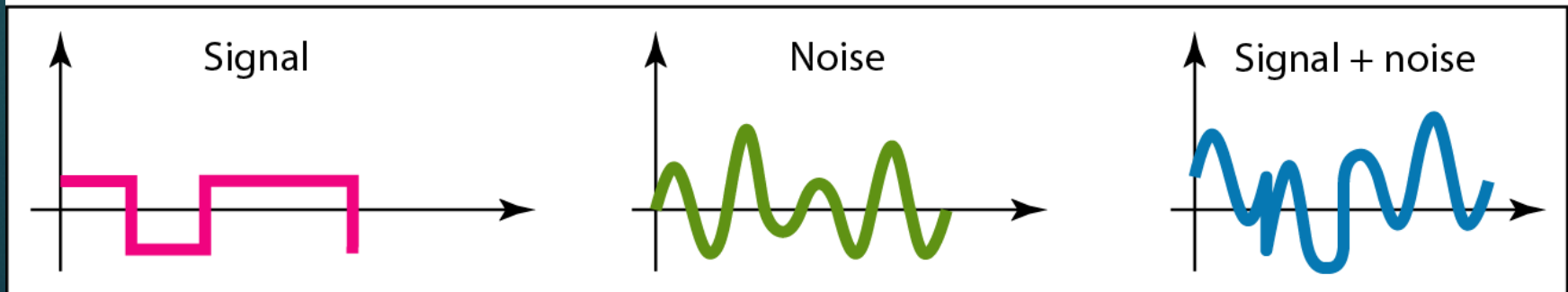
SNR = Average Signal power / Average Noise Power

- ▶ It is the ratio between two powers.
- ▶ It is usually given in dB and referred to as SNR_{dB} , defined as
 $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$

Figure 3.30 Two cases of SNR: a high SNR and a low SNR



a. Large SNR



b. Small SNR

3-5 DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Topics discussed in this section:

- Noiseless Channel: Nyquist Bit Rate
- Noisy Channel: Shannon Capacity
- Using Both Limits



Note

Increasing the levels of a signal increases the probability of an error occurring, in other words it reduces the reliability of the system.

Capacity of a System

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- ▶ The bit rate of a system increases with an increase in the number of signal levels we use to denote a symbol.
- ▶ A symbol can consist of a single bit or “n” bits.
- ▶ The number of signal levels = 2^n .
- ▶ As the number of levels goes up, the spacing between level decreases -> increasing the probability of an error occurring in the presence of transmission impairments.

Nyquist Theorem

- ▶ Nyquist gives the upper bound for the bit rate of a transmission system by calculating the bit rate directly from the number of bits in a symbol (or signal levels) and the bandwidth of the system (assuming 2 symbols/per cycle and first harmonic).
- ▶ Nyquist theorem states that for a **noiseless** channel:

$$C = 2 B \log_2 2^n$$

C= capacity in bps

B = bandwidth in Hz

Shannon's Theorem

- ▶ Shannon's theorem gives the capacity of a system in the presence of noise.

$$C = B \log_2(1 + \text{SNR})$$

In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second.



Note

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.